EX2

CRIANDO E RODANDO CONTAINER : NEO4J

```
davi@davi:~/neo4j_csv$ docker run -d \
  --name neo4j \
  -p 7474:7474 -p 7687:7687 \
  -e NEO4J_AUTH=neo4j/12345678 \
  -e NEO4J_dbms_allow__file__import__from__https__urls=true \
  neo4j:latest
Unable to find image 'neo4j:latest' locally
latest: Pulling from library/neo4j
4f4fb700ef54: Already exists
37fb86c2a767: Already exists
aac7585c8fdf: Download complete
6ed51959837c: Downloading [============>                                      ]  23.07MB/92.6MB
145c1cbb0967: Download complete
86f9220b6551: Downloading [================>                                  ]  77.59MB/257.1MB
d62687815a31: Download complete
```

EXERCÍCIO 2.1: Inserir dados obtidos a partir de um arquivo CSV
Acessando: davi@davi:~/neo4j_csv$ docker exec -i neo4j cypher-shell -u neo4j -p 12345678 << 'EOF'

pequena correção no container:

```
davi@davi:~/neo4j_csv$ docker run -d \
  --name neo4j \
  -p 7474:7474 -p 7687:7687 \
  -e NEO4J_AUTH=neo4j/12345678 \
  -e NEO4J_dbms_security_procedures_unrestricted=apoc.\* \
  neo4j:latest
0db92547a3d937c73fb330cd90c5f0926733385bdae30d36aa6cba42368c1b0e
davi@davi:~/neo4j_csv$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS         PORTS
                                                                 NAMES
0db92547a3d9   neo4j:latest   "tini -g -- /startup…"   4 seconds ago   Up 3 seconds   0.0.0.0:7474->7474/tcp, [::]:7474-
>7474/tcp, 7473/tcp, 0.0.0.0:7687->7687/tcp, [::]:7687->7687/tcp   neo4j
davi@davi:~/neo4j_csv$
```

```
davi@davi:~/neo4j_csv$ docker exec -it neo4j cypher-shell -u neo4j -p 12345678
Connected to Neo4j using Bolt protocol version 6.1 at neo4j://localhost:7687 as user neo4j.
Type :help for a list of available commands or :exit to exit the shell.
Note that Cypher queries must end with a semicolon.
neo4j@neo4j> LOAD CSV WITH HEADERS FROM 'https://raw.githubusercontent.com/neo4j-contrib/northwind-neo4j/master/data/sup
pliers.csv' AS row
             CREATE (s:Supplier {
               supplierID: row.SupplierID,
               companyName: row.CompanyName,
               contactName: row.ContactName,
               country: row.Country
             });
0 rows
ready to start consuming query after 65 ms, results consumed after another 0 ms
Created 29 nodes, added 29 labels
neo4j@neo4j>
```

```
neo4j@neo4j> LOAD CSV WITH HEADERS FROM 'https://raw.githubusercontent.com/neo4j-contrib/northwind-neo4j/master/data/cat
egories.csv' AS row
             CREATE (c:Category {
               categoryID: row.CategoryID,
               categoryName: row.CategoryName,
               description: row.Description
             });
0 rows
ready to start consuming query after 50 ms, results consumed after another 0 ms
Created 8 nodes, added 8 labels
neo4j@neo4j>
```

cadastrados, acessei o cli do neo com docker exec -it neo4j cypher-shell -u neo4j -p 12345678, preferivel
EXERCÍCIO 2 — Parte 2: Indexar cada nó a partir dos seus IDs

```
neo4j@neo4j> CREATE INDEX category_id_idx FOR (c:Category) ON (c.categoryID);
0 rows
ready to start consuming query after 19 ms, results consumed after another 0 ms
Added 1 index
neo4j@neo4j> CREATE INDEX product_id_idx FOR (p:Product) ON (p.productID);
0 rows
ready to start consuming query after 13 ms, results consumed after another 0 ms
Added 1 index
neo4j@neo4j>
```

EXERCÍCIO 2 — Parte 3: Transformar chaves estrangeiras em relacionamentos

```
davi@davi:~/neo4j_csv$ docker exec neo4j mkdir -p /var/lib/neo4j/import
davi@davi:~/neo4j_csv$ docker cp suppliers.csv neo4j:/var/lib/neo4j/import/
Successfully copied 4.23kB (transferred 6.14kB) to neo4j:/var/lib/neo4j/import/
davi@davi:~/neo4j_csv$ docker cp categories.csv neo4j:/var/lib/neo4j/import/
Successfully copied 2.46kB (transferred 4.1kB) to neo4j:/var/lib/neo4j/import/
davi@davi:~/neo4j_csv$ docker cp products.csv neo4j:/var/lib/neo4j/import/
Successfully copied 4.47kB (transferred 6.14kB) to neo4j:/var/lib/neo4j/import/
davi@davi:~/neo4j_csv$
```

com alguns locais:

```
neo4j@neo4j> CREATE
               (s1:Supplier {supplierID: "1", companyName: "Exotic Liquids", contactName: "Charlotte Cooper", country: "UK"}),
               (s2:Supplier {supplierID: "2", companyName: "New Orleans Cajun Delights", contactName: "Shelley Cain", country: "USA"}),
               (s3:Supplier {supplierID: "3", companyName: "Grandma Kelly's Homestead", contactName: "Regina Murphy", country: "USA"}),
               (s4:Supplier {supplierID: "4", companyName: "Tokyo Traders", contactName: "Yoshi Nagase", country: "Japan"}),
               (c1:Category {categoryID: "1", categoryName: "Beverages", description: "Soft drinks, coffees, teas, beers, and ales"}),
               (c2:Category {categoryID: "2", categoryName: "Condiments", description: "Sweet and savory sauces, relishes, spreads, and seasonings"}),
               (c3:Category {categoryID: "3", categoryName: "Confections", description: "Desserts, candies, and sweet breads"}),
               (c4:Category {categoryID: "6", categoryName: "Meat/Poultry", description: "Prepared meats"}),
               (c5:Category {categoryID: "7", categoryName: "Produce", description: "Dried fruit and bean curd"}),
               (c6:Category {categoryID: "8", categoryName: "Seafood", description: "Seaweed and fish"}),
               (p1:Product {productID: "1", productName: "Chai", unitPrice: 18.0, supplierID: "1", categoryID: "1"}),
               (p2:Product {productID: "2", productName: "Chang", unitPrice: 19.0, supplierID: "1", categoryID: "1"}),
               (p3:Product {productID: "3", productName: "Aniseed Syrup", unitPrice: 10.0, supplierID: "1", categoryID: "2"}),
               (p4:Product {productID: "4", productName: "Chef Anton's Cajun Seasoning", unitPrice: 22.0, supplierID: "2", categoryID: "2"}),
               (p5:Product {productID: "6", productName: "Grandma's Boysenberry Spread", unitPrice: 25.0, supplierID: "3", categoryID: "2"}),
               (p6:Product {productID: "9", productName: "Mishi Kobe Niku", unitPrice: 97.0, supplierID: "4", categoryID: "6"}),
               (p7:Product {productID: "10", productName: "Ikura", unitPrice: 31.0, supplierID: "4", categoryID: "8"});
0 rows
ready to start consuming query after 114 ms, results consumed after another 0 ms
Created 17 nodes, set 69 properties, added 17 labels
neo4j@neo4j> MATCH (p:Product), (s:Supplier)
             WHERE p.supplierID = s.supplierID
             CREATE (p)-[:SUPPLIED_BY]->(s);
0 rows
ready to start consuming query after 66 ms, results consumed after another 0 ms
Created 7 relationships
neo4j@neo4j> MATCH (p:Product), (c:Category)
             WHERE p.categoryID = c.categoryID
             CREATE (p)-[:PART_OF]->(c);
0 rows
ready to start consuming query after 38 ms, results consumed after another 0 ms
Created 7 relationships
neo4j@neo4j> MATCH (s:Supplier)<-[:SUPPLIED_BY]-(p:Product)-[:PART_OF]->(c:Category)
             RETURN s.companyName AS Fornecedor, collect(DISTINCT c.categoryName) AS CategoriasAtendidas;
+-----------------------------------------------------------+
| Fornecedor                  | CategoriasAtendidas         |
+-----------------------------------------------------------+
| "Exotic Liquids"            | ["Beverages", "Condiments"] |
| "Grandma Kelly's Homestead" | ["Condiments"]              |
| "New Orleans Cajun Delights" | ["Condiments"]             |
| "Tokyo Traders"             | ["Meat/Poultry", "Seafood"] |
+-----------------------------------------------------------+

4 rows
ready to start consuming query after 70 ms, results consumed after another 8 ms
neo4j@neo4j>
```

EXERCÍCIO 2.3 (Parte B) — Consulta com Patterns (`-->`)

```
neo4j@neo4j> MATCH (s:Supplier)<-[:SUPPLIED_BY]-(p:Product)-[:PART_OF]->(c:Category)
             RETURN s.companyName AS Fornecedor, collect(DISTINCT c.categoryName) AS CategoriasAtendidas;
+-----------------------------------------------------------+
| Fornecedor                  | CategoriasAtendidas         |
+-----------------------------------------------------------+
| "Exotic Liquids"            | ["Beverages", "Condiments"] |
| "Grandma Kelly's Homestead" | ["Condiments"]              |
| "New Orleans Cajun Delights" | ["Condiments"]             |
| "Tokyo Traders"             | ["Meat/Poultry", "Seafood"] |
+-----------------------------------------------------------+

4 rows
ready to start consuming query after 2 ms, results consumed after another 2 ms
neo4j@neo4j>
```

RELACIONAMENTO E CONSULTA

```
neo4j@neo4j> MATCH (p:Product), (s:Supplier)
             WHERE p.supplierID = s.supplierID
             CREATE (p)-[:SUPPLIED_BY]->(s);

             MATCH (p:Product), (c:Category)
             WHERE p.categoryID = c.categoryID
             CREATE (p)-[:PART_OF]->(c);
0 rows
ready to start consuming query after 4 ms, results consumed after another 0 ms
Created 7 relationships
0 rows
ready to start consuming query after 3 ms, results consumed after another 0 ms
Created 7 relationships
neo4j@neo4j> MATCH (s:Supplier)<-[:SUPPLIED_BY]-(p:Product)-[:PART_OF]->(c:Category)
             RETURN s.companyName AS Fornecedor, collect(DISTINCT c.categoryName) AS CategoriasAtendidas;
+-----------------------------------------------------------+
| Fornecedor                  | CategoriasAtendidas         |
+-----------------------------------------------------------+
| "Exotic Liquids"            | ["Beverages", "Condiments"] |
| "Grandma Kelly's Homestead" | ["Condiments"]              |
| "New Orleans Cajun Delights" | ["Condiments"]             |
| "Tokyo Traders"             | ["Meat/Poultry", "Seafood"] |
+-----------------------------------------------------------+

4 rows
ready to start consuming query after 2 ms, results consumed after another 2 ms
neo4j@neo4j>
```